

Monomial $f(q) = a \cdot q^6$ Report

Polynomial

$$f(q) = aq^6$$

Naive

$$g_{\text{naive}}(X) = ax^6$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^6 + 15aq^4\sigma^2 + 45aq^2\sigma^4 + 15a\sigma^6$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) &= 36a^2q^{10}\sigma^2 + 1170a^2q^8\sigma^4 + 12480a^2q^6\sigma^6 \\ &\quad + 49500a^2q^4\sigma^8 + 61020a^2q^2\sigma^{10} + 10170a^2\sigma^{12}\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) &= 36a^2q^{10}\sigma^2 + 1395a^2q^8\sigma^4 + 13830a^2q^6\sigma^6 \\ &\quad + 51975a^2q^4\sigma^8 + 62370a^2q^2\sigma^{10} + 10395a^2\sigma^{12}\end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 15aq^4\sigma^2 + 45aq^2\sigma^4 + 15a\sigma^6$$

Unbiased

$$g_{\text{unb}}(X) = -15a\sigma^6 + 45a\sigma^4x^2 - 15a\sigma^2x^4 + ax^6$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^6$$

$$\begin{aligned}\text{Var}(g_{\text{unb}}(X)) &= 36a^2q^{10}\sigma^2 + 450a^2q^8\sigma^4 + 2400a^2q^6\sigma^6 \\ &\quad + 5400a^2q^4\sigma^8 + 4320a^2q^2\sigma^{10} + 720a^2\sigma^{12}\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{unb}}) &= 36a^2q^{10}\sigma^2 + 450a^2q^8\sigma^4 + 2400a^2q^6\sigma^6 \\ &\quad + 5400a^2q^4\sigma^8 + 4320a^2q^2\sigma^{10} + 720a^2\sigma^{12}\end{aligned}$$

Comparison

$$\text{mean gap} = -15aq^4\sigma^2 - 45aq^2\sigma^4 - 15a\sigma^6$$

$$\begin{aligned}\text{variance gap} &= -720a^2q^8\sigma^4 - 10080a^2q^6\sigma^6 - 44100a^2q^4\sigma^8 \\ &\quad - 56700a^2q^2\sigma^{10} - 9450a^2\sigma^{12}\end{aligned}$$

$$\begin{aligned}\text{MSE gap} &= -945a^2q^8\sigma^4 - 11430a^2q^6\sigma^6 - 46575a^2q^4\sigma^8 \\ &\quad - 58050a^2q^2\sigma^{10} - 9675a^2\sigma^{12}\end{aligned}$$

$$\text{Bias}(g_{\text{naive}})^2 = 225a^2q^8\sigma^4 + 1350a^2q^6\sigma^6 + 2475a^2q^4\sigma^8 + 1350a^2q^2\sigma^{10} + 225a^2\sigma^{12}$$